Command line installation instructions

- On Mac or Linux, use Terminal
- On Windows, install gitbash from https://git-scm.com/downloads/win

Python installation instructions

- Install Python 3.8 or higher (3.12 recommended) from https://www.python.org/downloads/. If you already have Python installed, you can check your version by calling

```
python3 --version
```

- **Option A: install the required Python packages with the provided install_bio334_env.py script.**
    - Create a virtual environment with the required Python packages:

    ```
    python3 install_bio334_env.py
    ```

    - Activate the environment. **MacOS/Linux users, run this:**

    ```
    source bio334_env/bin/activate
    ```

    Windows users, run this:

    ```
    call bio334_env\Scripts\activate.bat
    ```

    - Test the environment. If the output of the command below is "True", the installation was successful:

    ```
    python3 -c "import cobra;import
    jupyterlab;print(int(cobra.__version__.split('.')[
    1]) > 4 and int(jupyterlab.__version__[0]) > 3)"
    ```

    - Please cut-and-paste the command and its output and send it to Mark: mark.robinson@mls.uzh.ch or via Slack
- **Option B: Follow the instructions below:**
    - Open your terminal. Try the following command:

    ```
    python3 -m venv bio334_env && source
    bio334_env/bin/activate
    ```

    - If the above fails, you will need to install venv or virtualenv. **MacOS/Linux users, use the following command:**

    ```
    sudo apt install -y python3-venv && python -m venv
    bio334_env && source bio334_env/bin/activate
    ```

    - **Windows users, run this instead:**

```
pip install virtualenv && virtualenv bio334_env &&
call bio334_env\Scripts\activate.bat
```

- Install pip in the virtual environment. **In Mac/Linux, use the following command:**

```
wget https://bootstrap.pypa.io/get-pip.py &&
python get_pip.py
```

- **On Windows**, open your browser and paste the following URL https://bootstrap.pypa.io/pip/pip.pyz
  It will download a zip file with a Python script that can be run directly with the following command:

```
python pip.pyz
```

- Install COBRApy in the virtual environment using pip. **In Mac/Linux, use the following command:**

```
bio334_env/bin/pip install cobra
```

- **On Windows**, run the following command instead:

```
bio334_env\Scripts\pip install cobra
```

- Test the environment. If the output of the command below is “True”, the installation was successful:

```
python3 -c "import cobra;import
jupyterlab;print(int(cobra.__version__.split('.')[
1]) > 4 and int(jupyterlab.__version__[0]) > 3)"
```

- Please cut-and-paste the command and its output and send it to Mark: mark.robinson@mls.uzh.ch or via Slack

R/RStudio installation instructions

- Install R 4.4.3 from the Swiss CRAN mirror: https://stat.ethz.ch/CRAN/
- Install RStudio from Posit: https://posit.co/download/rstudio-desktop/
- Run RStudio and cut-paste the following commands into the **Console** to install R/Bioconductor packages

```
install.packages("BiocManager")
BiocManager::install("SingleCellExperiment")
sessionInfo()
```

- Please cut and paste the output of the `sessionInfo()` command and send it to Mark Robinson: mark.robinson@mls.uzh.ch or via Slack